Activity 2.7 – Round Robin Arbiter

1.Design Approach:

Designed a four-requester round robin arbiter using Verilog HDL. Implemented priority rotation logic to provide fair access to a shared resource among multiple requesters.

2.Verification Steps:

- Created a separate testbench.
- Applied reset to initialize priority.
- Activated multiple request signals.
- Verified grant signal generation.
- Checked one-hot grant behaviour.
- Tested priority rotation.
- Simulated using ModelSim.

3.Results:

The testbench checks grant selection and priority rotation. PASS/FAIL messages are displayed.

4.Issues Faced:

Priority must rotate correctly after each grant. Only one requester should receive access at a time. Simultaneous requests require proper arbitration.

5.Conclusion:

Successfully designed and verified a four-requester round robin arbiter using Verilog HDL.